DMSHNet: Multiscale and Multisupervised Hierarchical Network for Remote-Sensing Image Change Detection

Pengcheng Liu, Panpan Zheng, and Liejun Wang

***Abstract*— Change detection (CD) is an increasingly popular research direction in the field of remote sensing (RS). With the rapid development of RS, the resolution of RS images is gradually improving, which also puts forward higher requirements for the generalization ability of CD models. Due to its excellent feature extraction capabilities, convolutional neural networks (CNNs) have achieved great success in CD tasks in the past. However, CNN cannot model remote context, which also limits its performance on CD. In contrast, Transformer does well in modeling remote contextual information. To make an end, in order to make full use of the advantages of both, we fully combine CNN and Transformer and propose multiscale and multisupervised hierarchical network for RS Image CD, called DMSHNet. First, we introduce multiple convolution kernels of different sizes in the Siamese encoder to help DMSHNet perceive information of different scales. Second, we develop the change information enhancement module (CIEM), which mainly generates differential features at different layers. CIEM uses simple absolute value subtraction and atrous convolution to highlight changing information and suppress unchanged information in different ranges. Third, we propose multisupervised and multiscale cascade module (DMSCM), which integrates multilayer differential features generated by CIEM on the basis of fully considering local information, global information, and multiscale information. We have conducted sufficient experiments on four public datasets, and the experimental results show that our DMSHNet achieves excellent performance. Our source code is available at https://github.com/ahlpc/DMSHNet.git.**

***Index Terms*— Change detection (CD), convolutional neural networks (CNNs), transformer.**

I. Introduction

IN THE field of remote sensing (RS), change detection (CD) is a challenging task and has become a popular research direction. CD is a technology that obtains surface state changes in the observation area by observing the same area at different times [1]. In other words, the CD task is to identify surface changes in accurately registered RS dual-phase images and predict change areas in images through binary mapping. Therefore, CD has wide applications in the fields of land use [2], disaster assessment [3], and urban expansion [4].

RS technology originated in the early 19th century [5]. With the rapid progress of science and technology, the development of RS technology has gone through three main stages of development: ground, air, and space. Meantime, RS images are being produced with increasingly higher resolution [6], and these images provide more useful information. At the same time, it also brings more formidable challenges to CD [7]. First, due to the characteristics of RS images such as wide width and large coverage area, the images collected under different weather [8], seasons [9], shooting angles [10], and other collection conditions are very different and may show different colors. These effects may cause the model to perceive changes in the environment beyond the changing targets. Second, multiresolution RS images place higher demands on the generalization of CD models [11].

In the development process of the field of RS CD, many experts and scholars have proposed many excellent algorithms [12], [13], [14], [15]. In the early stages, many algorithms emerged that relied on random forest [16], decision trees [17], and support vector machines [18]. Although traditional methods are easy to understand and implement and have higher accuracy when processing low- and medium-resolution RS images, they are not suitable for high-resolution RS image data.

In recent years, thanks to the rapid progress of big data and high-performance computing and other technologies, deep-learning artificial intelligence technology represented by convolutional neural networks (CNNs) and Transformer has achieved unprecedented development and has been widely used in many image processing tasks such as image classification [5], semantic segmentation [19], [20], and object detection [21], [22], [23], [24], it has achieved outstanding results in semantic segmentation and target detection. CNN can adaptively learn multilayer features in samples with the support of large and diverse data and has excellent nonlinear inductive learning capabilities in the face of complex problems. Therefore, many RS image CD methods based on CNN have been proposed and proved successful. However, CNN cannot consider the long dependencies between features due to the constraints of the convolution kernel, and can only establish the relationship between the internal information of the convolution kernel. To alleviate the above native problems of

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CNN, part of the work adds attention mechanisms or convolutional layers. Although these methods have certain effects, they still have disadvantages in long-distance context information modeling which is an important method in the CD field.

The self-attention mechanism proposed by Transformer [25], which is particularly popular at the moment, solves the above-mentioned native problems of CNN. Due to Transformer's powerful long-distance information modeling capabilities, many CD algorithms have been adopted and achieved impressive accuracy, such as SwinSUNet [26] and FTN [28]. Although Transformer achieves the accuracy that exceeds that of pure CNN networks in the RS image CD task, there are still rooms that can be improved. First of all, Transformer is different from CNN which can only use convolution kernels to interact in local areas. Vision transformer (ViT) [29] achieves global information interaction by dividing the image into a series of patches and then using self-attention to update the token. However, due to the high complexity caused by the number of tokens, self-attention still faces huge computational and parameter challenges in the task, especially when the input is a high-resolution image [30]. Besides, Transformer ignores local small granularity information when obtaining the global context information of the image, which will have an adverse impact on the CD task results.

After discussion, we know that CNN and Transformer are complementary in extracting long and short distance information dependencies. Therefore, we propose a multiscale and multisupervision hierarchical network called DMSHNet which integrates the advantages of CNN and Transformer to complete the CD task in this article. First, inspired by previous work [29], [31], CNN is used in the shallow layer of the encoder, and Transformer is used in the deep layer. By adopting different schemes at different layers, short-range dependency information can be fully extracted in the shallow encoder network, whereas long-range dependency information can be extracted in the deep encoder network. Furthermore, we develop a change information enhancement module (CIEM) so that change features can be obtained at different levels and multiple ranges. Third, in the decoder part, multisupervised and multiscale cascade module (DMSCM) is designed. Based on the same strategy as the encoder, it fuses multilayer features and supervises the output of the first two layers. This ensures that multilayer features, local features, and global features can be fully integrated in the decoding part. In total, the Siamese feature extraction encoder can fully extract the hidden dual-temporal RS image features under different environmental conditions. The introduction of multiple convolution kernels of different sizes in CNN improves the generalization of the model and can be applied to different resolution RS image CD tasks.

The main contribution of our work can be summarized as follows.

1) This article propose an RS image CD model named DMSHNet. DMSHNet is a multilevel network featuring a dual-branch Siamese encoder, encompassing both CNN and Transformer. It aims to simultaneously capture local, global, and multiscale information from bitemporal RS images.
2) To extract spatiotemporal difference information across various levels, we introduce CIEM to process features extracted by the encoder. By employing operations such as subtraction, absolute value, and dilated convolution, it efficiently learns difference feature maps.
3) In order to generate more accurate change prediction maps, we adopt the strategy of multilayer feature fusion, cascade, and multilayer supervision, and propose DMSCM.
4) We conduct extensive experiments on four public datasets (CDD, BCDD, SYSU-CD, and GZCD) with different resolutions to verify the performance of our proposed method. Experimental results on four datasets show that our method achieves better performance than the other eight methods.

II. Related Works

In this chapter, we will elaborate on several technical routes and their representative models in the field of CD. As shown in the Fig. 1, models based on CNN, Transformer, and CNN-Transformer, respectively.

A. CNN-Based Model

In the development process of CD, many experts and scholars have proposed a large number of pure CNN method models. Inspired by Unet, Daudt et al. [32] proposed CNN-based networks FC-EF, FC-Siam-conc, and FC-Siam-diff. FC-EF, FC-Siam-conc, and FC-Siam-diff all utilize a multilevel encoder based on CNN. Among them, FC-EF concatenates the images before feeding them into the network encoder, while FC-Siam-conc and FC-Siam-DIFF input the images into a weight-shared Siamese encoder based on CNN. Unlike FC-Siam-diff, which emphasizes calculating differences at the feature level, FC-Siam-conc, as shown in Fig. 1(a), merges or concatenates features at every layer. Fang et al. [33] proposed a pure CNN network SNUNet-CD. It uses a shared-weight CNN encoder to extract multilayer features of dual-temporal images. It enhances the capabilities of discovering the context information through adopting improved channel attention. The introduction of skip connections helps the multilayer features extracted by the encoder to be transferred to the encoder while maintaining granular information. Feng et al. [35] propose DMINet, which uses ResNet18(CNN) [21] to extract features from bitemporal images in the encoder and then uses a joint attention module to guide the attention distribution of the dual branches in the decoder. Features that pass through subtraction and concatenation operations are aligned and prediction is finally completed.

B. Transformer-Based Model

After Transformer emerges in the field of natural language, it becomes popular due to its powerful modeling capabilities. Since then, ViTs have also been widely used in the field of CD. Some researchers have also proposed CD models based on Transformer. Zhang et al. [26] proposed the first pure Transformer CD network named SwinSUNet. As shown in Fig. 1(b), SwinSUNet builds a U-shaped network based

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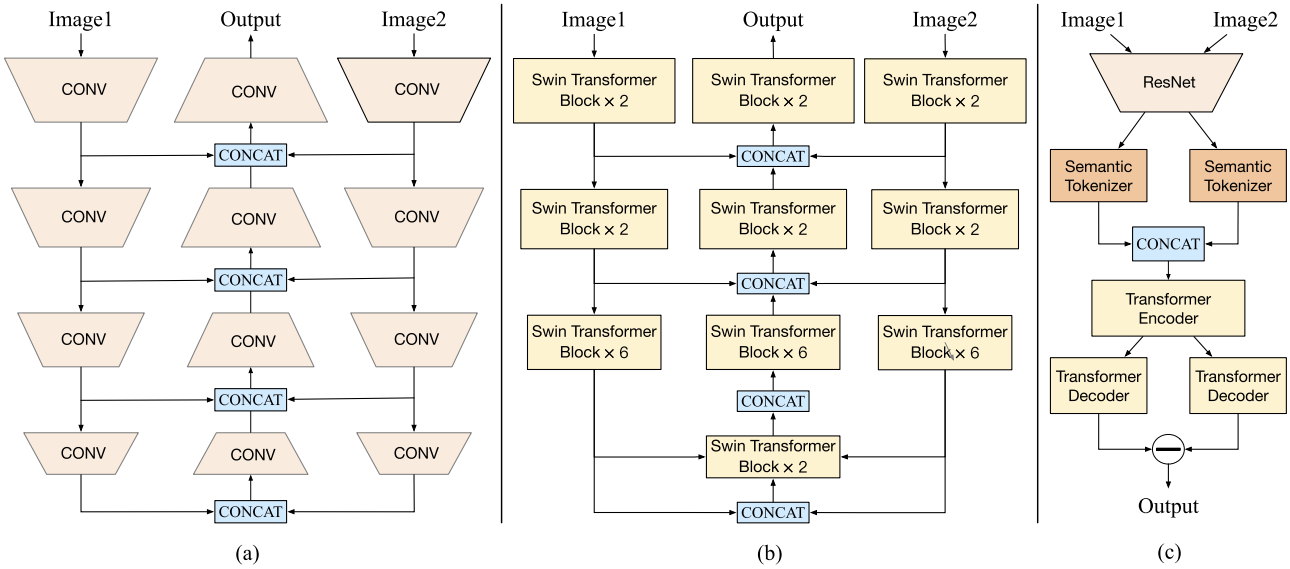

Fig. 1. Representative flowchart of CD networks. (a) FC-Siam-conc. (b) SwinSUNet. (c) BIT.

on Swin Transformer (Swin-T), which applies Swin-T to the encoder and decoder. Bandara and Patel [37] proposed ChangeFormer, which mainly uses a Siamese Transformer encoder network to extract the feature of bitemporal images and a lightweight multilayer perceptron (MLP) decoder to fuse multilevel feature differences. CAST has been proposed by Zhang et al. [27]. In ViT, the layer normalization is performed before the multihead self-attention (MSA) or MLP operations are performed, and finally, the residual connections are performed. In the encoder part of CAST, the MSA or MLP operation is performed first, and then the residual and layer normalization operations are performed. In the decoder part, the multihead cross-attention or MLP operation is performed first, and then the residual and layer normalization operations are performed.

C. CNN-Transformer-Based Model

The combination of Transformer and CNN can complement each other, so some researchers have also proposed some CD models based on CNN and Transformer. Chen et al. [36] propose the bitemporal image transformer (BIT), which for the first time uses a combination of CNN and Transformer to complete the CD task. In BIT, multilevel features of bitemporal images are extracted using ResNet, followed by feeding these multilevel features into Transformer for semantic enhancement to obtain refined features of the two images as shown in Fig. 1(c). Finally, prediction map is obtained through subtraction and projection operations. Song et al. [38] propose ACABFNet. In the encoder part, this model uses CNN and Transformer to extract local features and global features of dual-temporal spliced images, respectively. In the process of extracting multilayer features, the fusion of local features and global features is completed between each layer. In the decoder part, after fully fusing multilayer features, a CNN network is used to generate a prediction map. Tang et al. [39] propose WNet, which is similar to ACABFNet but different in that the WNet network does not splice the images before input. Instead, it uses CNN and Transformer to extract dual-phase image features respectively in the encoder and fuse the features at each layer in the decoder. Fu et al. [40] propose SLDDNet. Different from the previous network, WNet only fuses the multilayer features extracted by CNN and Transformer in the decoder part, while SLDDNet uses CNN and Transformer in each layer of the encoder. The local features extracted by CNN and the global features extracted by Transformer are fused in each layer. Feng et al. [41] propose ICIF-Net, which uses CNN and Transformer to process bitemporal images respectively in the encoder, and then fuses the features by CNN and Transformer in the decoder.

III. Methodology

In this chapter, we introduce our CD network named DMSHNet. Fig. 2 illustrates the complete framework of DMSHNet. For each pair of images to be detected, we first extract the local, global, and multilevel features of the images in the encoder by using both CNNs and transformers. Second, to capture and enhance the image change information, we introduce CIEM, which amplifies the difference information in the images by gradually reducing the receptive field. Third, to achieve multilevel feature fusion, optimize intermediate layers, and avoid the negative effects caused by deep supervision, we propose the DMSCM. This module implements full supervision at the outputs of the first and second layers of the network, which improves the overall performance and accuracy of CD. Finally, we will demonstrate the algorithmic flow of DMSHNet, as shown in Algorithm 1.

A. Multisupervised and Multiscale Cascade Module

1) Motivation: In the scenario of CD, traditional supervision often focuses on the final layer so that intermediate layers cannot be trained well [32], [43]. To address this issue, deep supervision is employed to reinforce the training of intermediate layers with extra supervised tasks on their final

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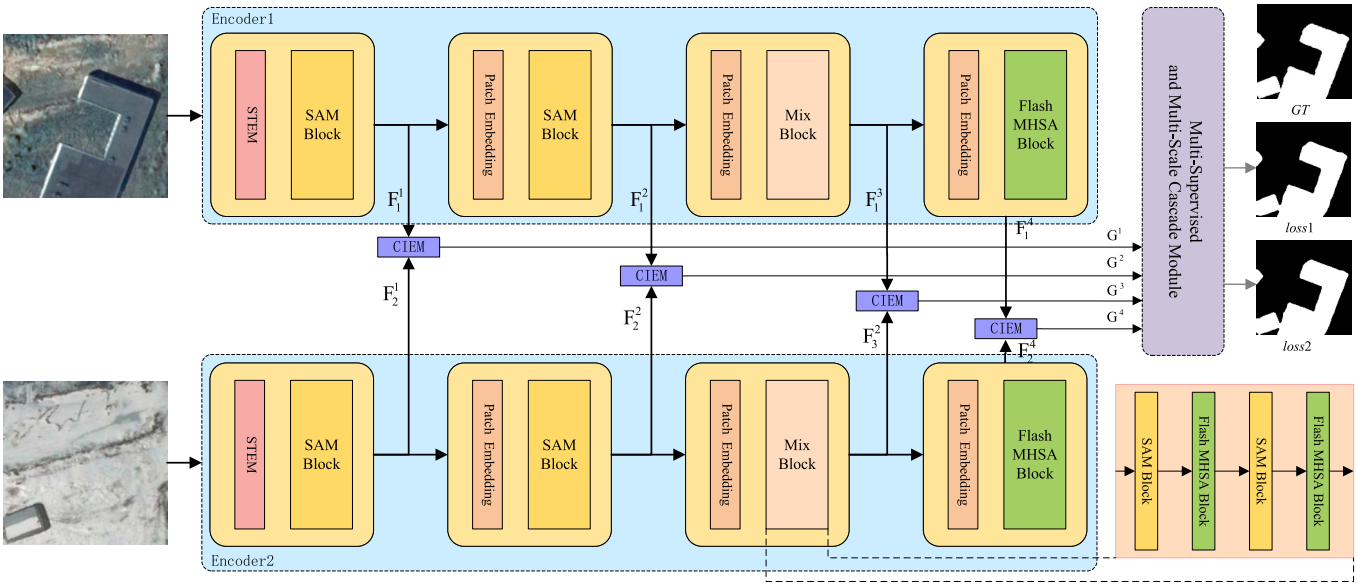

Fig. 2. Overall framework of DMSHNet.

Algorithm 1 DMSHNet

Input: $\mathcal{I} = \{(I_1, I_2)\}$ (a pair of patches)
Output: $\mathcal{M}_1, \mathcal{M}_2$ (change prediction maps)

// step1: extract multilayer features;
while $j = 1, 2$ **do**
  $F_j^1 = SAM\ Block(I_j)$
  $F_j^2 = SAM\ Block(F_j^1)$
  $F_j^3 = Mix\ Block(F_j^2)$
  $F_j^4 = Flash\ MHSA\ Block(F_j^3)$
end
// step2: get and enhance change information
while $k = 1, 2, 3, 4$ **do**
  $G^k = CIEM(F_1^k, F_2^k)$
end
// step3:fuse multilayer features and predict changes
$\mathcal{M}_1, \mathcal{M}_2 = DMSCM(G^1, G^2, G^3, G^4)$

outputs [28]. It is observed that, however, fully deep supervision often mitigates the capturing of low-level features [44].

To curb this issue, the discussion above brings us to one intuitive idea naturally, combining the traditional and deep supervision together and only performing the full-scale supervision on the shallow layers. Meanwhile, a bridge layer is required to transform the information smoothly between low-level features and high-level semantic ones. Specifically, in order to fully integrate the semantic information and detailed information in the multilayer difference features of RS dual-temporal images, we adopt the Transformer To CNN architecture from the bottom–up to adapt to the change of semantic information to detailed information in the process of the network changing from deep to shallow.

To this end, we develop a partially deep supervision model, in which half of the output layers are supervised, and we expect the proposed approach is not only applicable for seizing the low-level features but also high-level semantic ones.

2) Multisupervised and Multiscale Cascade Module: We propose DMSCM, as shown in Fig. 3. In DMSCM, we achieve bottom–up path information fusion. Following the idea of FPN, feature maps generate with the same spatial size at the same stage of the network, where each feature level corresponds to a processing stage. At each processing stage of DMSCM, we perform different operations. In the deep layer of DMSCM, we utilize Flash Attention to process features; at the next level, a mixed structure of scale-aware modulation (SAM) and Flash Attention is employed to achieve the transition from semantic features to low-level features; at the shallow level of DMSCM, SAM is utilized to suppress background noise while also providing full-size supervision for the shallow-level outputs.

3) Scale-Aware Modulation: To minimize the negative effects caused by background noise, we implement SAM [42] in the network's shallow layer. SAM can highlight foreground objects and filter out background information.

a) Multihead mix convolution: Tocapture spatial features at different scales, we adopt multihead mix convolution (MHMC). As shown in the Fig. 4, the input features are divided into four heads along the channel after passing through the linear layer, and depth-separable convolutions with different convolution kernel sizes are performed on each head. The convolution kernel size starts from 3 and increases from left to right until the convolution kernel size is 9. Thus, MHMC can be formulated as follows:

$$\mathrm{MHMC}(X) = \mathrm{Concat}\big(\mathrm{DW}_{k_1\times k_1}(X_1), \mathrm{DW}_{k_2\times k_2}(X_2), \ldots, \mathrm{DW}_{k_n\times k_n}(X_n)\big) \quad (1)$$

where $k_i \in \{3, 5, 7, \ldots, K\}$ means the size of the kernel that increases by 2 for each adjacent head, and DW() denotes

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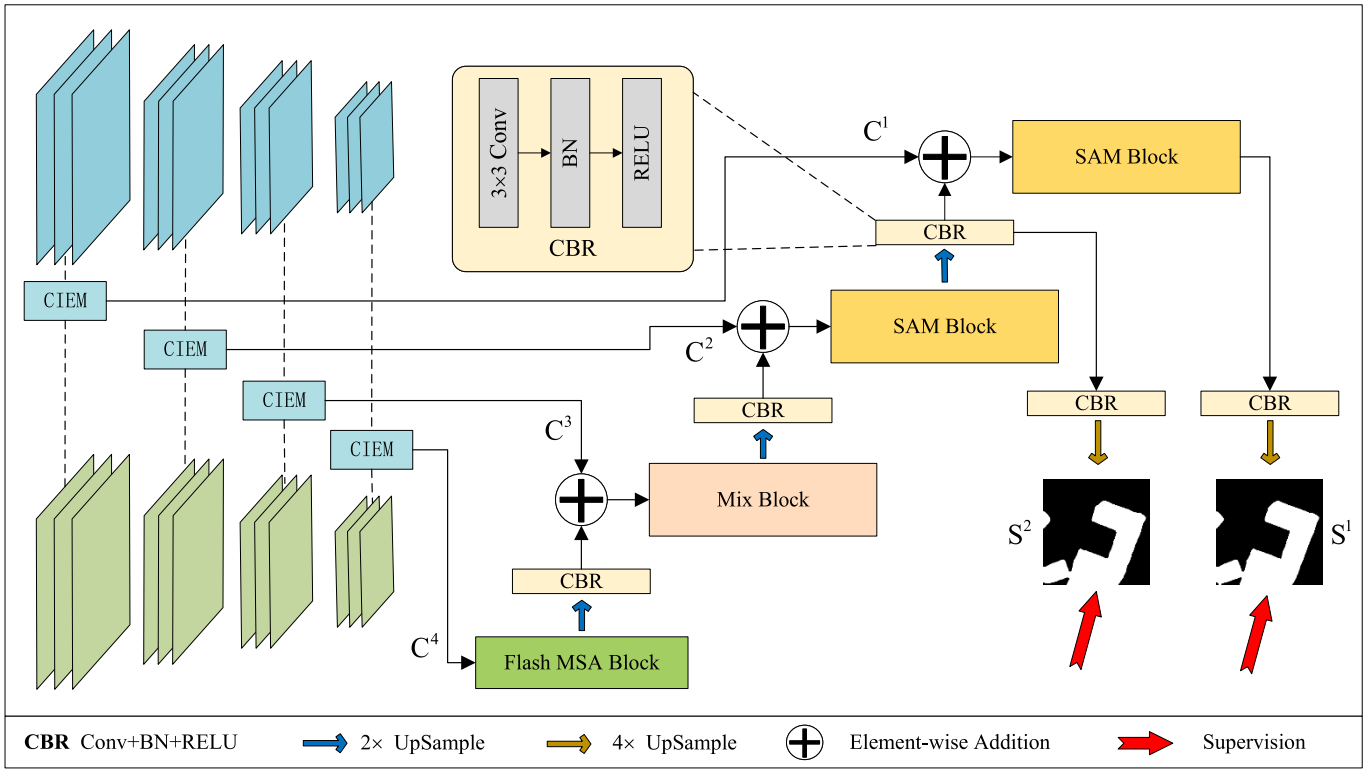

Fig. 3. Overall structure of DMSCM.

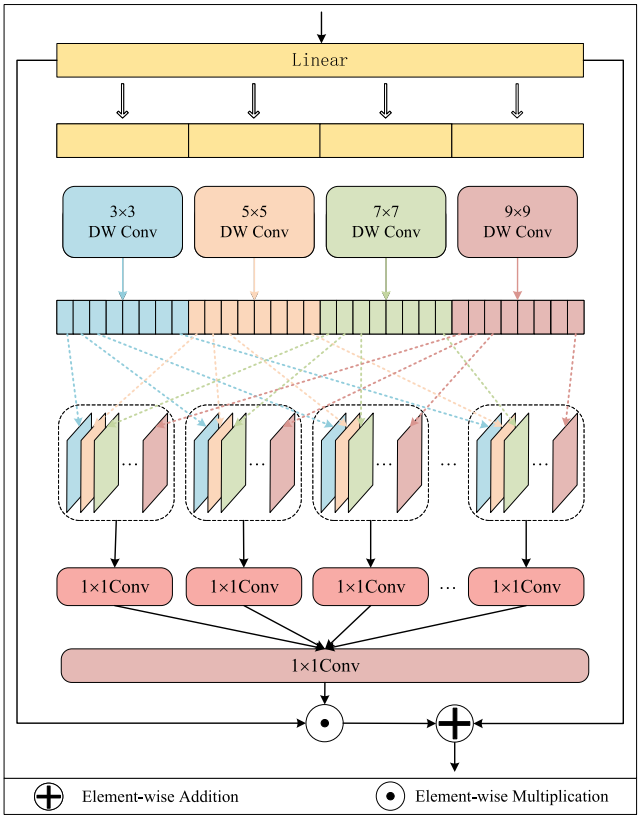

Fig. 4. Overall structure of SAM.

depthwise convolution, and Concat() means that concat the results of depthwise convolution operations.

b) Scale-aware aggregation: In MHMC, there is no information interacts between heads, so scale-aware aggregation (SAA) is introduced. The procedures of SAA involve a sequence of grouping, shuffling, and fusion. During the grouping phase, the multihead features produced by MHMC are grouped based on the rules: Groups $= C/\text{Heads_num}$. C represents the number of channels of input feature X. In the shuffling stage, we sequentially select one channel from each head to create new groups. Next, within each group, feature fusion occurs using a 1×1 convolutional layer. Then, feature fusion between heads involves merging groups and passing the combined results through another 1×1 convolutional layer. Thus, the process of SAA is as follows:

$$M = W_{\text{inter}}([G_1, G_2, \ldots, G_P]) \tag{2}$$

$$G_i = W_{\text{intra}}\left(\left[H_1^i, H_2^i, \ldots, H_Q^i\right]\right) \tag{3}$$

$$H_j^i = \text{DWConv}_{k_j \times k_j}\left(x_j^i\right) \in R^{H \times W \times 1} \tag{4}$$

where W_{inter} and W_{intra} are weight matrices of point-wise convolution. P and Q denote the number of groups and the number of heads, respectively. $H_j \in \mathbb{R}^{H \times W \times C}$ denotes the jth head with depth-wise convolution, and H_j^i means the ith channel in the jth head.

After obtaining multiscale features using MHMC and fusing them by SAA, we can get an output feature called M. Then, we adopt scalar product operation to modulate the input features X which passes through linear layers. Thus, for the input feature $X \in \mathbb{R}^{\mathbb{H} \times \mathbb{W} \times \mathbb{C}}$, the entire SAM process can be

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formulated as follows:

$$M = \mathrm{SAA}(\mathrm{MHMC}(WX)) \tag{5}$$

$$\mathrm{SAM}(X) = X + M \odot WX \tag{6}$$

where W denotes weight matrices of linear layers. $\odot$ denotes the element-wise multiplication.

4) Flash MHSA Block: The Transformer architecture relies on self-attention, which incurs quadratic time and storage complexity with respect to the sequence length. As a result, processing long sequences can be slow and memory-intensive. While existing acceleration schemes focus on reducing attentional computation and memory requirements, they overlook memory accesses [45], [46], [47]. Flash Attention [49] stands out from previous acceleration schemes by minimizing memory reads and writes during attention calculation.

Inspired by Flash Attention's significant speedup over traditional Transformers in NLP, we apply it to RS image CD. Flash Attention's core involves block computations in self-attention matrix multiplications. A challenge lies in implementing block-wise softmax calculations during self-attention.

The block calculation of softmax is depicted by the formula below

$$f\left(x^{(j)}\right) = \left[e^{x_1^{(j)} - m\left(x^j\right)}, e^{x_2^{(j)} - m\left(x^{(j)}\right)}, \ldots, e^{x_n^j - m\left(x^{(j)}\right)}\right] \tag{7}$$

$$\mathrm{softmax}\left(x^{(j)}\right) = \frac{f\left(x^{(j)}\right)}{\sum_1^n f\left(x^{(j)}\right)_i} \tag{8}$$

where j denotes the jth block of input vector X and $m(x^{(j)})$ represents the max value of the element in $x^{(j)}$. By using (7) and (8), the softmax value of the jth block can be determined. Next, we aim to obtain the maximum value from the entire vector X. Thus, we need to save the global values $m_{\max}$ and l_{all}. After processing $X^{(j+1)}$ is done, the above global values are depicted by the formula below

$$l'_{\mathrm{all}} = e^{m_{\max} - m'_{\max}} l_{\mathrm{all}} + e^{m\left(x^{(j+1)}\right) - m'_{\max}} l\left(x^{(j+1)}\right) \tag{9}$$

where $l(x^{(j+1)})$ means the sum of the elements of $f(x^{(j+1)})$ computed element-wise and $m'_{\max}$ denotes the max values between $x^{(j)}$ and $x^{(j+1)}$

$$\mathrm{softmax}'\left(x^{(j+1)}\right) = \frac{\mathrm{softmax}\left(x^{(j+1)}\right) l\left(x^{(j+1)}\right) e^{m\left(x^{(j+1)}\right)} - m'_{\max}}{l'_{\mathrm{all}}} \tag{10}$$

where $\mathrm{softmax}(x^{(j+1)})$ and $\mathrm{softmax}'(x^{(j+1)})$, respectively, represent the local softmax value of the $(j+1)$th block of input vector X and the global softmax values of the first $j+1$ blocks of input vectors X.

5) DMSCM Structure: In DMSCM, we perform multilayer feature fusion on the multilayer differential features generated by CIEM and generate effective prediction maps. We denote the multilevel feature maps passed into DMSCM as $C^i (i \in (1, 2, 3, 4))$, where i represents the input feature map at the ith stage. And DMSCM can be expressed by using mathematical formulas as follows:

$$\begin{aligned} C^{4'} &= \mathrm{Conv}\left(\mathrm{Up}\left(\mathrm{Flash\ MHSA\ Block}\left(C^4\right)\right)\right) \\ C^{3'} &= \mathrm{Conv}\left(\mathrm{Up}\left(\mathrm{Mix\ Block}\left(C^{4'} + C^3\right)\right)\right) \\ C^{2'} &= \mathrm{Conv}\left(\mathrm{Up}\left(\mathrm{SAM\ Block}\left(C^{3'} + C^2\right)\right)\right) \\ C^{1'} &= \mathrm{SAM\ Block}\left(C^{2'} + C^1\right) \end{aligned} \tag{11}$$

where Mix Block represents the combined module of cross-stacked SAM Block and Flash MHSA Block, as shown in Fig. 2

$$\begin{aligned} S^{2'} &= \mathrm{Up}\left(\mathrm{Conv}\left(C^{2'}\right)\right) \\ S^{1'} &= \mathrm{Up}\left(\mathrm{Conv}\left(C^{1'}\right)\right) \end{aligned} \tag{12}$$

where $S^{1'} \in \mathbb{R}^{256\times256\times3}$ and $S^{2'} \in \mathbb{R}^{256\times256\times3}$ represents the first and second layer predict map of DMSCM, respectively. It is worth mentioning that we still take the predict map $S^{1'}$ of the first layer as the final prediction map shown in the visualization Section IV-C.

6) Remarks: In terms of network output supervision, our approach differs from previous works that employed deep supervision (e.g., FTN [28] and SLDDNet [40]) and shallow supervision (e.g., BIT [36], FC-Siam-CONC [32], SNUNetCD [33], SwinSUNet [26], and ChangeFormer [37]). We supervise the outputs of the first and second layers of the network. This strategy aids in better training the intermediate layers of the network and aligns with the principle that shallow networks learn low-level features while deep networks learn semantic features.

Although some previous works have adopted the idea of fusing CNN and Transformer, the main difference in our work lies in our respect for the changes in dependency relationships from shallow to deep layers within the network. Therefore, we use CNNs to extract low-level features in the shallow layers of the network and Transformers to extract semantic features in the deep layers. Additionally, to accommodate the changes in long-range dependencies from shallow to deep layers, we use a Mix Block that stacks CNN and Transformer. BIT [36] first extracts features using a CNN network and then enhances these features with a Transformer. WNet [39] Separately extracts features from bitemporal images using CNN and Transformer, then feeds these features into a CNN-based decoder to perform CD. SLDDNet [40] utilizes CNN and Transformer to extract bitemporal image features at each layer, merging global and local features within the layer before passing them to the next layer.

B. Change Information Enhancement Module

1) Motivation: There do exist some state-of-the-art works for measuring the subtle changes between two images, while most of them fall prey to the impact of noise and falsely report changes due to the limitation of direct pixel-wise subtraction [36], [41], [48]. Aiming to a high detection accuracy, instead of a plain measurement, our proposed approach calls for a signal enhancement component that can amplify these subtle changes. Formally, we are looking for a transformation

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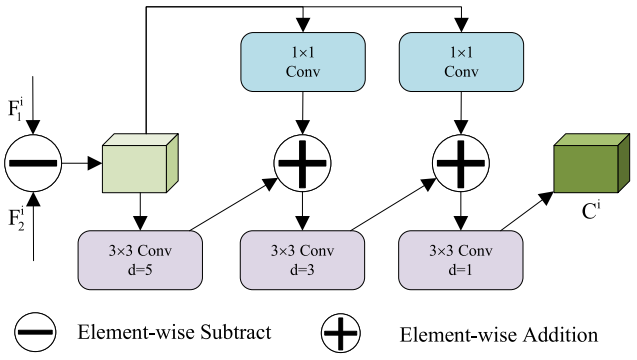

Fig. 5. Overall structure of CIEM.

function $f: X \to Y$ where X is the input space and Y refers to the output space, such that small changes in X result in much larger changes in Y.

Intuitively, one way to achieve this kind of signal amplification is through nonlinear functions that exhibit steep gradients around the region of interest. Specifically, in this work, we come up with one CIEM that takes use of dilated convolution networks to construct the nonlinear transformation function f.

2) Change Info Enhancement Module Structure: CIEM consists of a series of atrous convolutions and residual connections. First, it takes the absolute value of the result after subtracting the features of the same layer. Then, multiple convolutions and residual connections are employed to learn the differences as shown in Fig. 5. Using mathematical formulas, the main processes of CIEM can be expressed as follows:

$$\begin{aligned} G^i &= \mathrm{ABS}\left(F_1^i - F_2^i\right) \\ C_3^i &= \mathrm{Conv}_{1\times1}\left(G^i\right) + \mathrm{Conv}_{3\times3}^{p=5,d=5}\left(G^i\right) \\ C_2^i &= \mathrm{Conv}_{1\times1}\left(G^i\right) + \mathrm{Conv}_{3\times3}^{p=3,d=3}\left(C_3^i\right) \\ C^i &= \mathrm{Conv}_{3\times3}^{p=1,d=1}\left(C_2^i\right) \end{aligned} \tag{13}$$

where G^i denotes the absolute value of the subtraction of two features at the same level of the ith layer, and $C_j^i (i \in (1,2,3,4), j \in (1,2,3))$ means that the multiple convolution processed result of G^i, p and d denote the size of padding and dilation, respectively. And C^i means the final output of CIEM at the ith layers.

3) Remarks: Unlike previous works, our method explores the efficacy of change information by transitioning from a broad to a narrow scope. This hierarchical approach ensures a thorough examination of changes at various levels of detail, enhancing the accuracy and robustness of the detection process. In contrast, existing methodologies (e.g., FC-Siam-CONC [32], BIT [36], ChangeFormer [37], and SwinSUNet [26]) typically employ simpler techniques such as subtraction or concatenation of features before directly passing the outcomes to a decoder. These conventional methods might overlook subtle but critical changes, as they do not progressively refine the search for change information across different scales. Our approach, by methodically narrowing the focus, captures a more comprehensive set of change features, leading to more precise and reliable CD results.

C. Loss Function

In the task of CD, the number of pixels in the changed area is significantly less than the number of pixels in the unchanged area, resulting in an imbalance in the number of samples. To alleviate the impact of this problem, we combine cross-entropy loss and dice loss functions to train and optimize our network DMSHNet. During the network training process, $\mathcal{L}_{\mathrm{ce}}$ can treat any pixel equally, and $\mathcal{L}_{\mathrm{dice}}$ can pay attention to changing pixels. Therefore, the combination of $\mathcal{L}_{\mathrm{ce}}$ and $\mathcal{L}_{\mathrm{dice}}$ can balance changed and unchanged pixels to a certain extent. The cross-entropy and dice loss functions can be expressed as follows:

$$\mathcal{L}_{\mathrm{ce}} = -\frac{1}{N}\sum_{i=1}^{N}\left[Y_i \log S_i + (1-Y_i)\log(1-S_i)\right] \tag{14}$$

$$\mathcal{L}_{\mathrm{dice}} = 1 - \frac{2\sum_{i=1}^{N} S_i Y_i}{\sum_{i=1}^{N}(S_i + Y_i) + \epsilon} \tag{15}$$

$$\mathcal{L} = \mathcal{L}_{\mathrm{ce}} + \mathcal{L}_{\mathrm{dice}} \tag{16}$$

where ϵ value equals $1e-7$, and the reason for adding it is to avoid that the denominator is zero. N means the total amount of pixels. Y and S represent the Ground Truth map and the binary change prediction map, respectively. The whole loss function $\mathcal{L}$ used in DMSHNet can be formulated as (16).

IV. Experiment

A. Datasets

In this chapter, we conduct extensive experiments on four public datasets to verify the effectiveness of DMSHNet.

1) Change Detection Dataset: The change detection dataset (CDD) dataset contains 11 pairs of images from Google Earth with resolutions ranging from 3 to 10 cm per pixel [50]. The changes to be detected in the CDD dataset include changes in cars, roads, and buildings. In our experiment, the original image was cropped into 16 000 256 × 256 pixels image pairs, of which 10 000 images were used as the training set and the verification set and test set each had 3000 image pairs. The ratio of changing pixels to unchanging pixels in the CDD dataset is 1:7.19.

2) Building CDD: The building CDD (BCDD) dataset is a dataset that only detects changes in buildings [51]. The images of this dataset are mainly taken from aerial images before and after the 6.3 magnitude earthquake in a New Zealand city in March 2011. The original dataset image size is 32 507 × 15 354 pixels and the resolution of the image is 0.2 m per pixel. According to the previous method [51], in our subsequent experiments, we cropped it into 7484 pictures with a size of 256 × 256 pixels, and finally randomly assigned 5985 pictures as the training set, and the verification set and the test set included 794 and 795 pictures, respectively. The ratio of changing pixels to unchanging pixels in the BCDD dataset is 1:22.08.

3) Sun Yat-Sen University Dataset: The Sun Yat-Sen University Dataset (SYSU) dataset contains 800 pairs of aerial images of Hong Kong, and its detection types mainly include buildings, roads, and vegetation [53]. Its original dataset mainly includes 800 pictures of 1024 × 1024 pixels

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and the spatial resolution of its images is 0.5 m/pixel. In our experiments, the original images were cropped into images with a size of 256 × 256 pixels, and the training set, validation set, and test set included 12 000, 4000, and 4000 images, respectively. The ratio of changing pixels to unchanging pixels in the SYSU-CD dataset is 1:3.58.

4) Guangzhou Datasets: Guangzhou datasets (GZCD) uses the Google Earth service to collect 19 very-high-resolution image pairs with a spatial resolution of 0.55 m/pixel [52]. The data collection location was in the suburbs of Guangzhou, China. The 19 images vary in size from 1006 × 1168 to 4936 × 5224 pixels. In our experiments, we divided these images of different sizes into 3130 pairs of 256 × 256 images. According to the existing division rules, the training set, verification set, and test set contained 2504, 313, and 313 images, respectively. The ratio of changing pixels to unchanging pixels in the GZCD dataset is 1:9.40.

B. Settings of Experiments

1) Details of Implementation: During the training process of our model, the framework used is Pytorch and the graphics accelerator card used is Nvidia Tesla T4 with 16 GB of video memory. The optimization method in the experiment uses Adamw, and other key parameters are set as follows. The total number of epochs is 200. For data enhancement, horizontal flipping, vertical rotation, and rotation operations were used in the experiment. During the training process, after each epoch is trained on the training set, it is verified on the verification part of the dataset, and the model with the highest $F1$ on the verification set is saved. In the final testing phase, the previously saved model is selected and tested on the test set. The initial learning rate is 1e-4, which will gradually decrease as epochs increase. The updated rules of the learning rate are as follows:

$$\text{newlr} = \text{baselr} * \left(1 - \frac{\text{CurEpoch}}{\text{TotalEpoch}}\right)^{0.9} \tag{17}$$

where baselr, newlr, CurEpoch, and TotalEpoch denotes the initial learning rate, the updated learning rate, the current epoch number, and the number of total epochs.

2) Evaluation Standards: We used a total of five indicators to evaluate the experimental results, namely $F1$-Score (F1), Precision (Pre), Recall (Rec), overall accuracy (OA), and intersection over union (IoU). The above standards can be obtained using the following formula:

$$\text{Pre} = \text{TP}/(\text{TP} + \text{FP}) \tag{18}$$

$$\text{Rec} = \text{TP}/(\text{TP} + \text{FN}) \tag{19}$$

$$\text{IoU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{20}$$

$$\text{OA} = (\text{TP} + \text{TN})/(\text{TP} + \text{FP} + \text{FN} + \text{TN}) \tag{21}$$

$$F1 = 2/\left(\text{Recall}^{-1} + \text{Precision}^{-1}\right) \tag{22}$$

where TN, TP, FN, and FP denote the number of true negative, true positive, false negative, and false positive, respectively.

C. Comparisons Experiments

We choose the current nine CD methods for comparison, including FC-EF, FC-Siam-diff, FC-Siam-conc, SNUNet, SwinSUNet, BIT, ICIF-Net, ChangerFormer, and ACABFNet. Among them, the first four are based on CNN, the last four are based on CNN and Transformer, and only SwinSUnet is based on Transformer. All comparison methods selected for testing in this section use the original codes provided by the author. The experimental environment is consistent with that mentioned in Section IV-B.

1) Evaluation Indicator Results: The evaluation index results of different models on four public datasets are shown in Table I, where the first and second evaluation index results are highlighted in bold and underlined, respectively. It can be easily seen that in most cases our proposed DMSHNet outperforms others.

Taking CDD as an example, DMSHNet achieved improvements of 1.79%, 0.29%, and 1.73% in IoU, OA, and $F1$ compared to ACABFNet. Taking SYSU-CD as an example, DMSHNet achieved improvements of 10.34%, 1.51%, and 7.35% in IoU, OA, and $F1$ compared to ChangeFormer.

Experiments on four datasets with different resolutions can demonstrate that our method is effective in handling CD tasks with different image resolutions. Possible reasons for the positive experimental results include the following. First, we leveraged their respective advantages by using a combination of CNN and Transformer, which helped DMSHNet extract multiscale information, local information, and global information in RS images. Second, CIEM can highlight and enhance the changing information of dual-temporal images. Third, DMSCM ensures the quality of the final change map by using multilayer output supervision and fully integrating global information and local information.

2) Visualization Results: We randomly selected three pictures for each model result for visual result analysis. Please see the Fig. 6. Different colors in the picture represent different meanings. The white, black, red, and green tables represent TP, TN, FP, and FN. DMSHNet reduces the impact of different environmental conditions when processing RS images of different environmental conditions. Especially in situations where there is a large gap between images and environments, such as ice and snow coverage and overexposure, our DMSHNet still achieves better detection results. Second, DMSHNet can generate images that are closer to ground truth (GT) images. DMSHNet can complete the recognition of small target changes. In the fourth row of the Fig. 6, other methods generate a large number of FN pixels. Only our method completely detects the changes in small buildings, and the generated prediction map is very close to the GT map. In addition, our DMSHNet can still obtain good prediction results when detecting complex geographical environments. In the second row of the figure, where roads and buildings are arranged irregularly, our method produces the best prediction results. Furthermore, our method achieves excellent results when dealing with different change types, such as vegetation, ships, and buildings.

D. Efficiency Comparison

We select all the algorithms in the comparative experiment for efficiency analysis. Table II shows the parameters (Param.) and floating point operations per second (FLOPs) of each model. When counting various data in the table, the input

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TABLE I
COMPARISONS IN TERMS OF MIoU, F1, OA, REC, AND PRE ON FOUR DATASETS

Datasets		FC-EF [32]	FC-Siam-diff [32]	FC-Siam-conc [32]	BIT[36]	SNUNet [33]	SwinSUnet [26]	ACABFNet [38]	Change Former [37]	ICIF-Net[41]	DMSHNet (ours)
CDD	Pre	74.97	86.21	82.05	96.44	95.99	96.63	96.69	95.06	95.92	98.35
	Rec	68.64	72.27	74.49	92.27	95.51	91.39	97.08	95.33	95.11	98.86
	OA	93.59	95.36	95.07	98.69	99.01	98.35	99.26	98.87	98.95	99.55
	IoU	55.84	64.78	64.05	89.23	91.92	87.73	93.96	90.84	91.41	97.25
	F1	71.66	78.62	78.09	94.31	95.79	93.46	96.88	95.20	95.51	98.61
BCDD	Pre	83.35	62.13	52.77	83.21	84.88	92.73	94.00	91.45	89.73	98.59
	Rec	79.15	83.90	84.91	80.74	86.23	93.22	91.24	77.20	86.36	93.31
	OA	98.29	96.86	96.75	98.34	98.64	99.34	99.32	98.60	98.90	98.21
	IoU	68.34	55.51	48.25	69.43	74.75	86.87	86.22	72.01	78.59	92.08
	F1	81.19	71.39	65.09	81.96	85.56	92.97	92.60	83.73	88.01	95.88
SYSU-CD	Pre	76.47	87.80	88.89	82.15	81.16	84.51	81.27	78.81	78.85	90.91
	Rec	79.64	63.09	57.40	73.43	71.13	71.65	77.66	75.32	76.24	81.68
	OA	89.42	89.23	88.26	89.97	89.30	90.22	90.51	89.40	89.57	90.91
	IoU	63.97	58.01	53.56	63.33	61.05	63.34	65.87	62.63	63.29	72.97
	F1	78.02	73.42	79.76	77.54	75.82	77.55	79.42	77.02	77.52	84.37
GZCD	Pre	78.68	75.81	75.56	89.09	84.73	91.43	87.95	85.26	88.74	84.82
	Rec	73.92	70.73	74.02	80.15	81.30	89.12	85.94	70.96	84.97	99.80
	OA	95.24	94.66	94.85	96.94	95.56	98.02	97.33	95.74	97.34	97.11
	IoU	61.58	57.71	59.72	72.99	70.90	82.25	76.88	63.20	76.70	84.68
	F1	76.22	73.18	74.78	84.38	82.98	90.26	86.93	77.45	86.81	91.71

The best and second best results are highlighted in red and blue, respectively.

TABLE II
COMPARISON ON EFFICIENCY

Models	Param(M)	FLOPs(G)	Epoch Time(min)
FC-EF	1.35	3.58	0.78
FC-Siam-diff	1.35	4.73	0.77
FC-Siam-conc	1.55	5.33	0.77
BIT	3.50	10.63	2.03
SNUNet	12.03	54.83	5.77
SwinSUNet	40.95	12.02	8.05
ACABFNet	102.32	28.29	3.53
ChangeFormer	33.61	213.15	17.35
ICIF-Net	23.84	25.41	6.84
DMSHNet	33.26	149.04	2.42

The Epoch Time refers to the time required for each method to run an Epoch on the training set part of GZCD.

image size of each model is $3 \times 256 \times 256$. Since the DMSHNet we proposed uses Siamese encoder branches to extract multilayer features of dual-temporal RS images, this allows our DMSHNet to obtain more sufficient features but increases the number of parameters. In addition, because of using FlashAttention to replace the standard self-attention mechanism, helps DMSHNet to accelerate the training process and reduce access times of HBM.

The number of parameters of our model is slightly lower than ChangeFormer, less than ChangeFormer and SwinSUNet, and higher than BIT, SNUNet, and ICIF-Net. In terms of FLOPs, our model is lower than ChangeFormer and higher than BIT, SNUNet, SwinSUNet, and ACABFNet. In terms of training time, our model is only slower than BIT, but faster than SNUNet, SwinSUNet, ACABFNet, ChangeFormer, and ICIF-Net. It is worth mentioning that we use FlashAttention in the model to replace the standard self-attention mechanism in Transformer, which helps our model improve the training speed. However, the till operation of FlashAttention increases the amount of calculations. In general, our model outperforms ChangeFormer in terms of the number of parameters, FLOPs, and training speed, but we still need to continue to study and explore the potential of reducing parameters and improving training speed.

E. Ablation Study

To examine each module's impact on CD task performance extensively, we conduct meticulous ablation experiments in this Section IV-B. In addition, the experimental environment and key parameter settings of all experiments in this section are the same as those in Section IV-B.

We verified the effectiveness of each module in DMSHNet by removing modules one by one in ablation experiments. The ablation experiment results are shown in the Table III, where "w/o" and "w" are the abbreviations of "without" and "with", respectively, "CIEM" is the change information proposing module, "DMSCM" is the multiscale multisupervised cascade module, and "VanillaDecoder" is a simple Decoder composed of upsampling and concatenation. "DMSCM*" and "vanillaDecoder*" represent only supervising the top-level output of the decoder (top-level output means $S^{1'}$ in Fig. 3).

1) Effectiveness of CIEM: To accurately evaluate the effectiveness of CIEM across the entire model, we use a simple subtraction operation to replace CIEM. As shown in the Table III, after removing CIEM, the detection accuracy of IoU, OA, and $F1$ on CDD of our model dropped by 2.84%, 0.33%,

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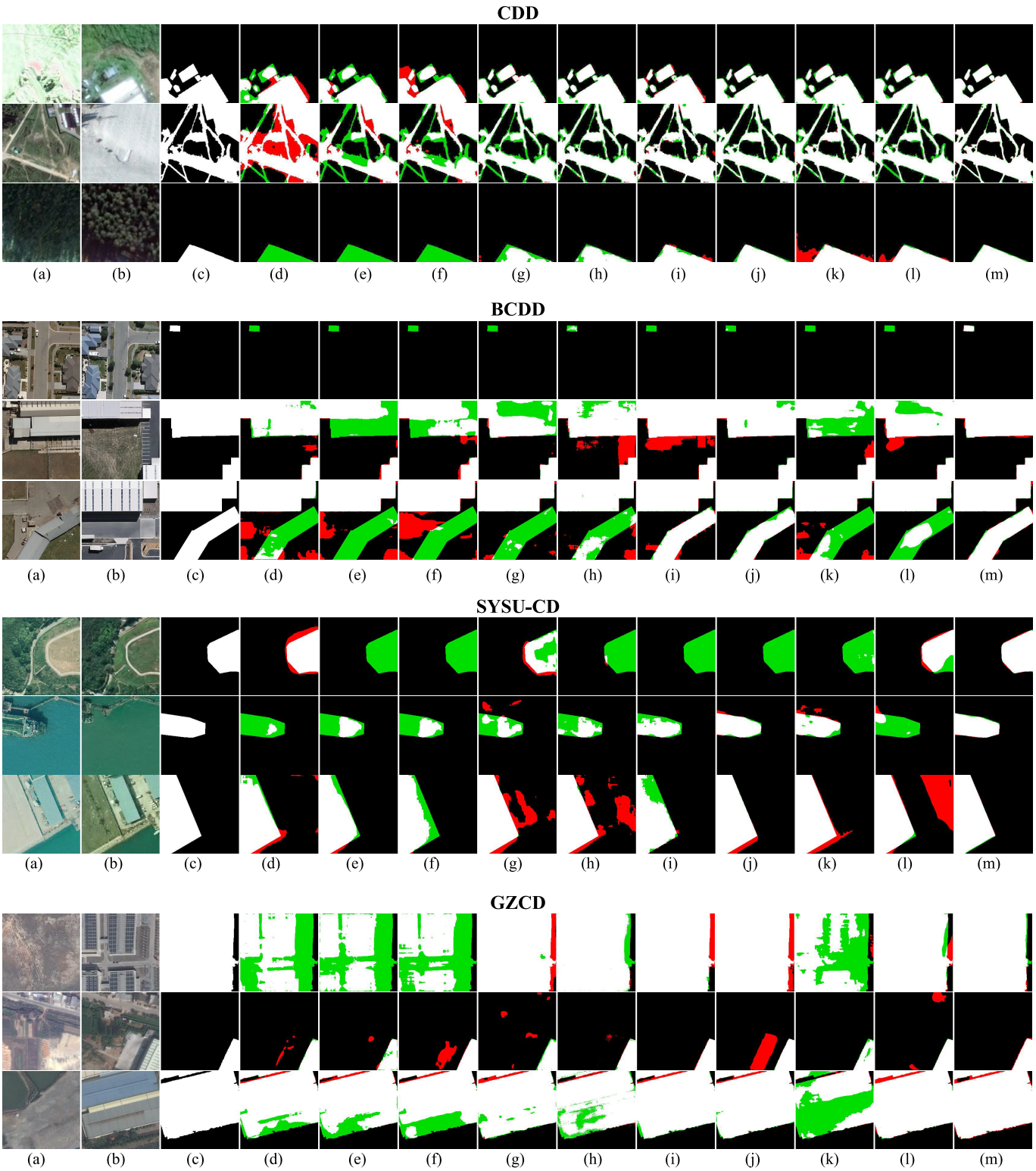

Fig. 6. Change maps of three pairs of bitemporal images randomly selected by different models in the test set of CDD, BCDD, SYSU-CD, and GZCD. White, black, red, and green pixels represent TP, TN, FP, and FN. (a) T1 image. (b) T2 image. (c) Ground truth. (d) FC-EF. (e) FC-Siam-diff. (f) FC-Siam-conc. (g) BIT. (h) SNUNet. (i) SwinSUNet. (j) ACABFNet. (k) ChangeFormer. (l) ICIF-Net. (m) DMSHNet.

and 2.85%, respectively. This represents different sized ranges which are very important to enhance change information.

2) Effectiveness of DMSCM: When verifying the effectiveness of DMSCM in the model, we designed a simple module called VanillaDecoder to replace DMSCM, which consists of basic upsampling and connection operations. As shown in the Table III, the detection accuracy of the replaced model on CDD dropped by 2.66%, 0.13%, and 1.80% compared to the original model. This illustrates the excellent performance of DMSCM in fusing multilayer features.

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TABLE III
ABLATION STUDIES ON THE COMPONENTS OF DMSHNet

Datasets	CDD			BCDD			SYSU-CD			GZCD		
	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1
a.w/o CIEM b.w DMSCM*	90.14	99.05	94.81	85.19	99.23	92.01	65.37	87.54	79.06	82.23	96.54	90.25
a.w/o CIEM b.w DMSCM	91.87	99.22	95.76	87.06	**99.32**	93.08	68.78	88.74	81.50	83.21	96.78	90.84
a.w CIEM b.VanillaDecoder*	93.72	99.40	97.76	84.66	99.20	91.69	66.71	89.23	80.03	81.52	96.38	89.82
a.w CIEM b.VanillaDecoder	93.82	99.41	96.81	82.71	99.10	90.54	66.28	89.31	79.72	82.38	96.58	90.34
a.w CIEM b.w DMSCM*	94.01	99.43	96.91	86.24	99.28	92.61	67.73	89.23	80.76	83.97	96.95	91.28
a.w CIEM b.w DMSCM	**97.25**	**99.55**	**98.61**	**92.08**	98.21	**95.88**	**72.97**	**90.91**	**84.37**	**84.68**	**97.11**	**91.71**

TABLE IV
ABLATION STUDIES ON DIFFERENT ENCODER OF DMSHNet

Network	CDD			BCDD			SYSU-CD			GZCD		
	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1
DMSHNet-C	94.72	99.23	97.29	84.54	97.49	91.62	67.50	89.98	80.59	81.69	96.42	89.92
DMSHNet-T	84.64	99.26	91.68	85.74	97.71	92.32	68.54	89.53	81.33	82.15	96.54	90.20
Ours	**97.25**	**99.55**	**98.61**	**92.08**	**98.21**	**95.88**	**72.97**	**90.91**	**84.37**	**84.68**	**97.11**	**91.71**

3) Effectiveness of Multisupervise: After verifying the effectiveness of CIEM and DMSCM, we also conducted ablation experiments to verify the effectiveness of multisupervision effectiveness. In terms of experimental settings, we conduct single-layer supervision and two-layer supervision experiments on the original model and models that replaced CIEM and DMSCM, respectively. In the Table III, we can see that our proposed DMSHNet has a lead of 1.66%, 0.12%, and 1.70% in detection accuracy IoU, OA, and $F1$ on the CDD dataset compared to the single-supervised DMSHNet. It is worth noting that multiple comparative experiments in the table show that the detection accuracy of double-layer supervision is mostly higher than the detection accuracy of single-layer detection.

F. Comparison With Different Encoder

Our model uses a combination of CNN and Transformer in the encoder part, as shown in Fig. 2. Therefore, it is necessary to investigate the role of the use of only CNN or Transformer. To achieve this goal, we replace the encoder part of DMSHNet with equivalent modules. The network after the replacement of the CNN module with the Transformer module is called DMSHNet-T and the network after the replacement of the Transformer module with the CNN module is called DMSHNet-C.

As shown in the Table IV, the original DMSHNet outperforms DMSHNet-C and DMSHNet-T on four public datasets. The reasons for these positive results are twofold. First, DMSHNet fully exploits the advantages of CNN and Transformer, which is beneficial for the capture of local and global features of RS images. Second, as DMSHNet gradually deepens the network, the stacked architecture of CNN and Transformer simulates the changes in long-range dependencies from shallow to deep, allowing each block at each stage to better match the computational characteristics.

V. CONCLUSION

To fully incorporate both global and local features, we present a closely integrated CD model of CNN and Transformer called DMSHNet. The CIEM module achieves effective change information in large and small ranges, respectively, while DMSCM merges low-level features and high-level features. Four public datasets, namely CDD, BCDD, GZCD, and SYSU-CD, are employed for extensive experimentation. Comparative experiments demonstrate that our model achieves state-of-the-art (SOTA) performance when compared with other networks. A significant amount of resources are consumed in our bitemporal image feature extraction branches during training. Hence, our focus for future work will be on developing lightweight models.

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